IMPLEMNTATION OF THE MODE OF DES-ECB,CBC.CFB,OFB

AIM

To implement the four modes of **DES (Data Encryption Standard)—ECB, CBC, CFB, and OFB**—for encrypting and decrypting a given plaintext using a secret key.

ALGORITHM

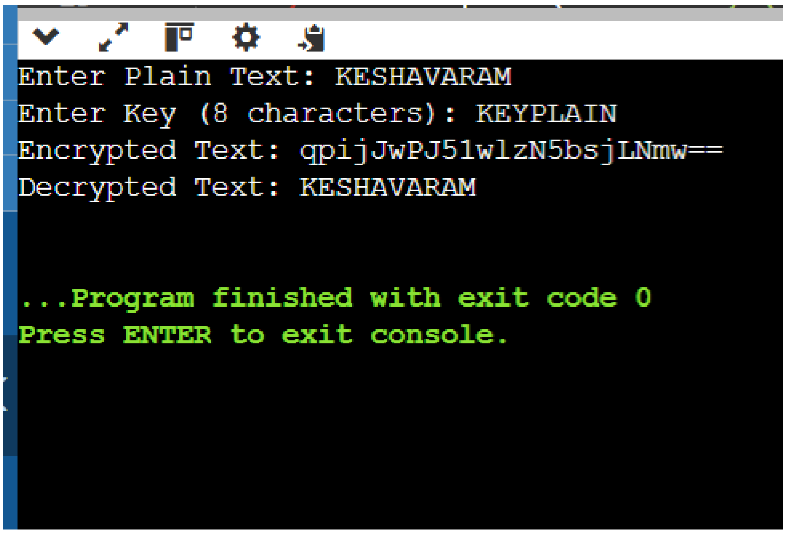
1. Start the program.
2. Read the plaintext and 8-byte DES key.
3. Select the required DES mode: **ECB, CBC, CFB, or OFB**.
4. Initialize the DES cipher with the selected mode and key.
5. Encrypt the plaintext and display the ciphertext.
6. Initialize the cipher in decryption mode using the same key and IV where required.
7. Decrypt the ciphertext and display the original plaintext.
8. Repeat the process for all four DES modes.
9. Stop the program.

PROGRAM:

(FOR ECB)

```
import javax.crypto.Cipher;
import javax.crypto.spec.SecretKeySpec;
import java.util.Base64;
import java.util.Scanner;

public class Main {
  public static void main(String[] args) throws Exception {
    Scanner sc = new Scanner(System.in);

    System.out.print("Enter Plain Text: ");
    String text = sc.nextLine();

    System.out.print("Enter Key (8 characters): ");
```

```
        String key = sc.nextLine();

        SecretKeySpec skey = new SecretKeySpec(key.getBytes(), "DES");
        Cipher cipher = Cipher.getInstance("DES/ECB/PKCS5Padding");

        cipher.init(Cipher.ENCRYPT_MODE, skey);
        byte[] encrypted = cipher.doFinal(text.getBytes());

        System.out.println("Encrypted Text: " +
            Base64.getEncoder().encodeToString(encrypted));

        cipher.init(Cipher.DECRYPT_MODE, skey);
        byte[] decrypted = cipher.doFinal(encrypted);

        System.out.println("Decrypted Text: " + new String(decrypted));

        sc.close();
    }
}
```

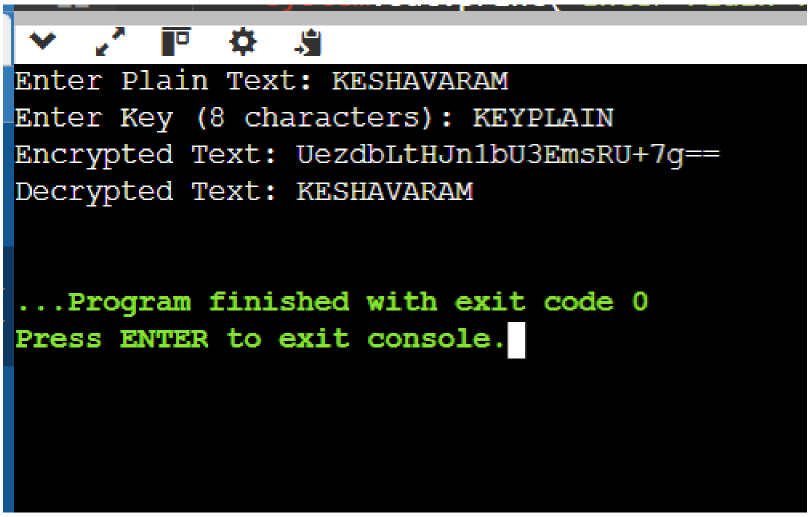


PROGRAM:

(CBC)

```
import javax.crypto.Cipher;
import javax.crypto.spec.IvParameterSpec;
import javax.crypto.spec.SecretKeySpec;
import java.util.Base64;
import java.util.Scanner;

public class Main {
    public static void main(String[] args) throws Exception {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Plain Text: ");
        String text = sc.nextLine();

        System.out.print("Enter Key (8 characters): ");
        String key = sc.nextLine();

        SecretKeySpec skey = new SecretKeySpec(key.getBytes(), "DES");
        IvParameterSpec iv = new IvParameterSpec("12345678".getBytes());

        Cipher cipher = Cipher.getInstance("DES/CBC/PKCS5Padding");

        cipher.init(Cipher.ENCRYPT_MODE, skey, iv);
        byte[] encrypted = cipher.doFinal(text.getBytes());

        System.out.println("Encrypted Text: " +
                Base64.getEncoder().encodeToString(encrypted));
```

```
        cipher.init(Cipher.DECRYPT_MODE, skey, iv);
        byte[] decrypted = cipher.doFinal(encrypted);

        System.out.println("Decrypted Text: " + new String(decrypted));

        sc.close();
    }
}
```

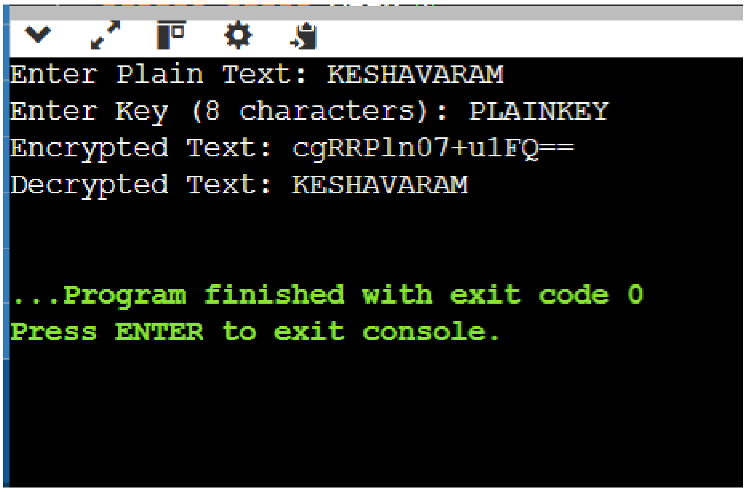


PROGRAM:

(CFB)

```
import javax.crypto.Cipher;
import javax.crypto.spec.IvParameterSpec;
import javax.crypto.spec.SecretKeySpec;
import java.util.Base64;
import java.util.Scanner;

public class Main {
```

```java
    public static void main(String[] args) throws Exception {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Plain Text: ");
        String text = sc.nextLine();

        System.out.print("Enter Key (8 characters): ");
        String key = sc.nextLine();

        SecretKeySpec skey = new SecretKeySpec(key.getBytes(), "DES");
        IvParameterSpec iv = new IvParameterSpec("12345678".getBytes());

        Cipher cipher = Cipher.getInstance("DES/CFB/NoPadding");

        cipher.init(Cipher.ENCRYPT_MODE, skey, iv);
        byte[] encrypted = cipher.doFinal(text.getBytes());

        System.out.println("Encrypted Text: " +
                Base64.getEncoder().encodeToString(encrypted));

        cipher.init(Cipher.DECRYPT_MODE, skey, iv);
        byte[] decrypted = cipher.doFinal(encrypted);

        System.out.println("Decrypted Text: " + new String(decrypted));

        sc.close();
    }
}
```

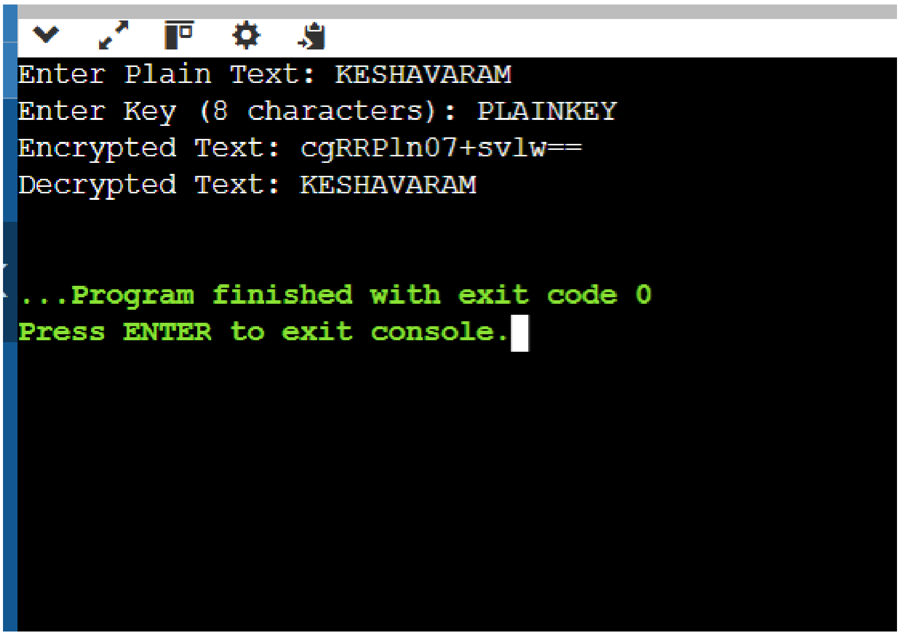


PROGRAM:

(OFB)

```
import javax.crypto.Cipher;

import javax.crypto.spec.IvParameterSpec;

import javax.crypto.spec.SecretKeySpec;

import java.util.Base64;

import java.util.Scanner;


public class Main {

    public static void main(String[] args) throws Exception {

        Scanner sc = new Scanner(System.in);


        System.out.print("Enter Plain Text: ");

        String text = sc.nextLine();


        System.out.print("Enter Key (8 characters): ");

        String key = sc.nextLine();


        SecretKeySpec skey = new SecretKeySpec(key.getBytes(), "DES");

        IvParameterSpec iv = new IvParameterSpec("12345678".getBytes());
```

```
        Cipher cipher = Cipher.getInstance("DES/OFB/NoPadding");

        cipher.init(Cipher.ENCRYPT_MODE, skey, iv);
        byte[] encrypted = cipher.doFinal(text.getBytes());

        System.out.println("Encrypted Text: " +
            Base64.getEncoder().encodeToString(encrypted));

        cipher.init(Cipher.DECRYPT_MODE, skey, iv);
        byte[] decrypted = cipher.doFinal(encrypted);

        System.out.println("Decrypted Text: " + new String(decrypted));

        sc.close();
    }
}
```



RESULT

Thus, the **four DES modes—ECB, CBC, CFB, and OFB—were implemented successfully**, and the original plaintext was recovered correctly after decryption.